Linear Algebra – Problem Set 6 — Solutions

Eigenvalues & Eigenvectors · Full solutions · 6 short problems · Original problems in the spirit of MIT 18.06 (Strang)

Every eigenvector below is checked by multiplying it out, and every spectrum is checked against tr and det. Those two habits catch almost every mistake in this topic.

Problem 1

$A = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$; test three vectors; why $\mathbf{v} \neq \mathbf{0}$.

(a)

$$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = (-1)\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

So $(1,0)$ has eigenvalue 3 and $(0,1)$ has eigenvalue -1. $(1,1)$ is not an eigenvector: $(3,-1) = c(1,1)$ would need $c = 3$ and $c = -1$ simultaneously. Eigenvectors: $(1,0)$ with $\lambda = 3$, and $(0,1)$ with $\lambda = -1$. **(b)** Because $A\mathbf{0} = \mathbf{0} = \lambda\mathbf{0}$ holds for *every* λ. If $\mathbf{0}$ counted, every number would be an eigenvalue of every matrix: there would be no finite list of n eigenvalues, and $\det A$ = product of eigenvalues, $\operatorname{tr} A$ = sum of eigenvalues would both be meaningless. Requiring $\mathbf{v} \neq \mathbf{0}$ is what makes the spectrum a real piece of information.

Problem 2

$B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: eigenvalues without a characteristic polynomial.

(a) Multiply the two symmetric directions:

$$B\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad B\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$\lambda = 3$ with direction $(1,1)$, and $\lambda = 1$ with direction $(1,-1)$. **(b)** A 2×2 matrix has a characteristic polynomial of degree 2, so it has exactly two eigenvalues counting multiplicity — and the two we found are distinct, so there is no room for a third. The audit confirms it: $\lambda_1 + \lambda_2 = 4 = \operatorname{tr} B = 2+2$, and $\lambda_1\lambda_2 = 3 = \det B = 4 - 1$.

Problem 3

$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$: the full computation.

(a) Only the diagonal changes when λI is subtracted:

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}, \qquad \det(A - \lambda I) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

$\lambda = 2$ and $\lambda = 5$. Audit: $2 + 5 = 7 = \operatorname{tr} A$, $2 \cdot 5 = 10 = \det A$ ✓. **(b)** Each eigenvalue gets its own reduction. $\lambda = 5$: $A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, so $x = y$ and $\mathbf{v} = t(1,1)$. $\lambda = 2$:

$A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \to \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$, so $2x + y = 0$, $y = -2x$ and $\mathbf{v} = t(1, -2)$. **(c)** Two multiplications:

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Both hold, so the pairing of eigenvalues and eigenvectors is correct.

Problem 4

$M = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$: the zero eigenvalue.

(a) Row 2 is twice row 1, so $\det M = 0$ and M is singular. A singular matrix has a non-zero vector with $M\mathbf{v} = \mathbf{0}$, i.e. an eigenvector with eigenvalue 0. The trace gives the other one: $\operatorname{tr} M = 1 + 4 = 5 = 0 + \lambda_2$, so $\lambda_2 = 5$. **(b)** For $\lambda = 0$, solve $M\mathbf{v} = \mathbf{0}$: $x + 2y = 0$ gives $\mathbf{v} = t(-2, 1)$. For $\lambda = 5$, $M - 5I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$ gives $y = 2x$, so $\mathbf{v} = t(1, 2)$. Check: $M(-2, 1) = (0, 0)$ ✓ and $M(1, 2) = (5, 10) = 5(1, 2)$ ✓. Eigenvalues 0 and 5; eigenvectors $t(-2, 1)$ and $t(1, 2)$.

Problem 5

$S = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$: repeated eigenvalue, one eigenvector.

(a) $S - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 0 & 2 - \lambda \end{bmatrix}$ is triangular, so $\det(S - \lambda I) = (2 - \lambda)^2$: $\lambda = 2$ with algebraic multiplicity 2. Now $S - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, and $(S - 2I)\mathbf{v} = \mathbf{0}$ says $y = 0$ with x free, so $\mathbf{v} = t(1, 0)$: the geometric multiplicity is 1. **(b)** No. Diagonalising a 2×2 matrix needs two independent eigenvectors to serve as the columns of P, and here the eigenspace is only the line $\operatorname{Span}\{(1, 0)\}$. Equivalently: if $S = PDP^{-1}$ then both diagonal entries of D would have to be 2, so $D = 2I$ and $S = 2I$ — false.

Problem 6

$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$.

(a) Column j of AP is A times column j of P; column j of PD is d_j times column j of P:

$$AP = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}, \qquad PD = \begin{bmatrix} 5 \cdot 1 & 2 \cdot 1 \\ 5 \cdot 1 & 2(-2) \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}.$$

They agree, so $AP = PD$. The equality is exactly the statement $A\mathbf{p}_1 = 5\mathbf{p}_1$ and $A\mathbf{p}_2 = 2\mathbf{p}_2$ — the columns of P are eigenvectors — which is why no inverse is needed to recognise a diagonalisation. **(b)** $(2, -1) = (1, 1) + (1, -2)$, and A multiplies the first piece by 5 and the second by 2 each time:

$$A^k \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 5^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2^k \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 5^k + 2^k \\ 5^k - 2^{k+1} \end{bmatrix}.$$

At $k = 1$: $(5 + 2,\ 5 - 4) = (7, 1)$, and indeed $A(2, -1) = (8 - 1,\ 4 - 3) = (7, 1)$ ✓.

Spotted an error, or a solution that skips a step? Open an issue on LA-hw and quote the problem number.